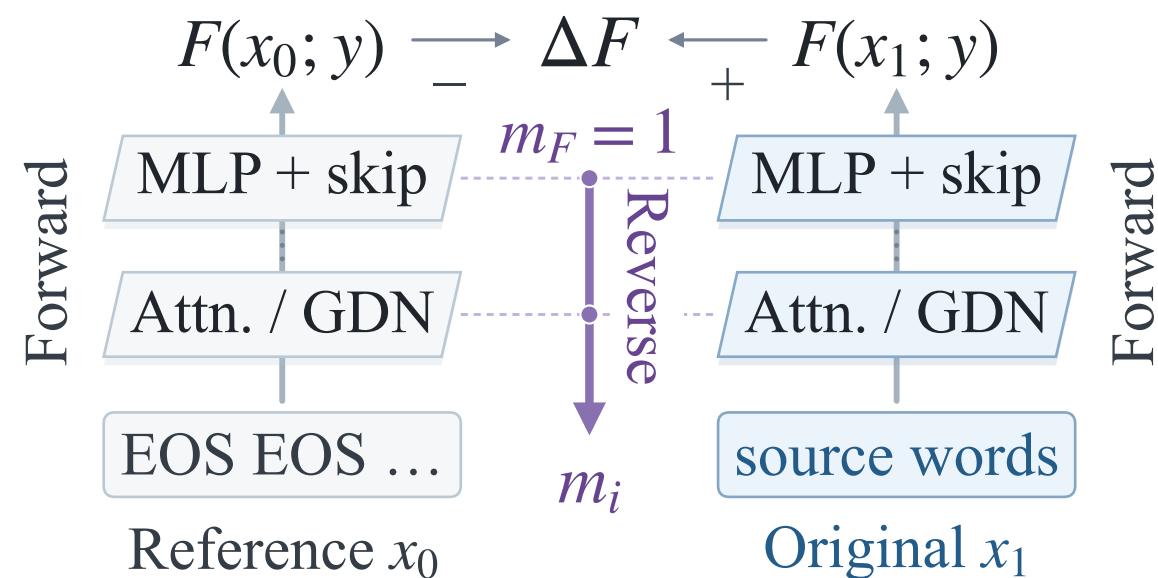


(a) From input change to credit

1 Two forward runs; fixed response y



2 **Local rules**
from both runs

3 **Reverse**

propagate coefficients m

4 **Token score**

$$\Delta h = D_f \Delta a$$

$$m_a = D_f^\top m_h$$

$$A_i = \langle m_i, \Delta e_i \rangle$$

(b) Attention

Select and carry

$$\begin{matrix} \text{3x3 grid} \\ P \end{matrix} \times \begin{matrix} \text{2x2 grid} \\ V \end{matrix} = \begin{matrix} \text{3x2 grid} \\ Y \end{matrix}$$

$$\bar{P} = (P_0 + P_1)/2, \quad \bar{V} = (V_0 + V_1)/2$$

Reverse

$$\Delta Y = \boxed{\bar{P} \Delta V} + \boxed{\Delta P \bar{V}}$$

Content Selection

Derive reverse rules

$$M_V = \bar{P}^\top M_Y \quad M_P = M_Y \bar{V}^\top$$

(c) Gated DeltaNet

Store and retrieve



Content

Control

Stored values

Memory use

Reverse

Input token scores

(d) Choose the playwright, not the composer.

Qwen3.5, MoreHopQA example 1

CONTEXT EXCERPTS

... William Shakespeare's play

... The score is by William Walton.

REQUESTED ROLE (EXCERPT)

... the author of the play ...



Full input

Name-span total (nats)

+1.56

-1.13

Token contribution (nats)

≤ -2.5

0

$\geq +2.5$

Shown spans underlined

Answer

William Shakespeare

$$\sum_i A_i = \Delta F$$

Across all input sources

(a) Qwen3-8B

Context

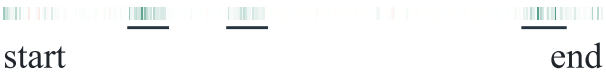
One of the special magic numbers for deafening-opium is: 5443951.

...
One of the special magic numbers for mammoth-barber is: 8698256.

Question

What are all the special magic numbers for deafening-opium, and mammoth-barber mentioned in the provided text?

Full input



(b) Qwen3.5-9B

Context

One of the special magic numbers for deafening-opium is: 5443951.

...
One of the special magic numbers for mammoth-barber is: 8698256.

Question

What are all the special magic numbers for deafening-opium, and mammoth-barber mentioned in the provided text?

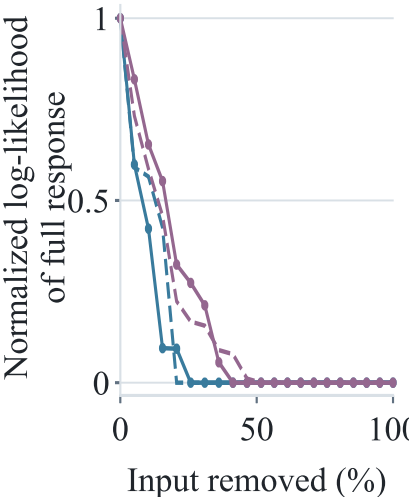
Full input



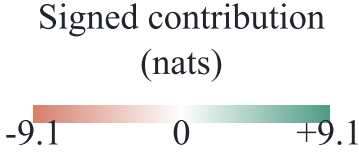
(c) Deletion

Answer

5443951, 8698256



- Qwen3-8B DT
- Qwen3-8B FT
- Qwen3.5-9B DT
- Qwen3.5-9B FT



(a) Qwen3-8B

Context

Townsend Putnam Coleman
III (born May 28, 1954) is
an American voice actor
...
he also did additional
voices in films "Fantasia
2000" (1999) and "Sing"
(2016)
...
It was directed and
written by Garth Jennings,
co-directed by Christophe
Lourdelet,

Question

How many consonants are
there in the last name of
the person who directed
and wrote the 2016 film
featuring the voice of
Townsend Coleman?

Full input



(b) Qwen3.5-9B

Context

Townsend Putnam Coleman
III (born May 28, 1954) is
an American voice actor
...
he also did additional
voices in films "Fantasia
2000" (1999) and "Sing"
(2016)
...
It was directed and
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Question

How many consonants are
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Townsend Coleman?

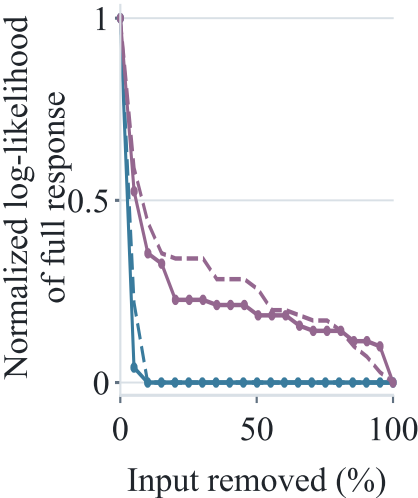
Full input



(c) Deletion

Answer

6 consonants
in Jennings



- Qwen3-8B DT
- Qwen3-8B FT
- Qwen3.5-9B DT
- Qwen3.5-9B FT

Signed contribution
(nats)



(a) Recall (%)

HotpotQA[‡]
Full-support
30-100

VT-H10[‡]
30-100

VT-H6[†]
20-100

VT-H4
10-80

VT-H2
20-100

MQ-Q2
10-80

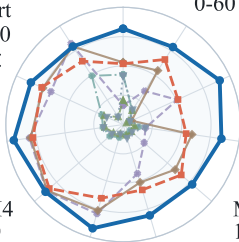
MQ-Q4
0-60

MQ-Q8
0-15

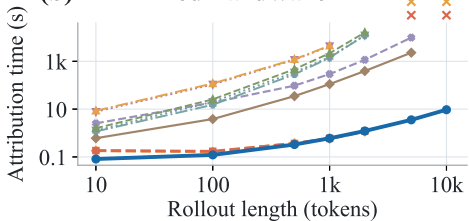
MV-V2
10-90

MV-V4
10-70

MV-V8
5-30



(b) Mixed hardware



DT FT FT-mh IGxAttn.
Perturb. REAGENT CLP OOM
IFR AttnLRP IG

Budgets: 10%; [†]20%; [‡]30%.

■ Dense finite propagation ■ Tiled finite propagation

